GARRETT DROLET

MACHINE LEARNING ENGINEER | AI ENGINEER

Texas | 817-673-0303 | garrettdrolet@gmail.com | LinkedIn | Portfolio

PROFESSIONAL SUMMARY

Machine learning/AI engineer and M.S. Computer Engineering graduate specializing in deep learning, time-series classification, reinforcement learning, and predictive modeling. Builds reproducible Python and MATLAB workflows spanning data preparation, model design, validation, evaluation, and technical communication. Combines applied AI expertise with four years of experience developing analytical and automation solutions for engineering and operational environments.

TECHNICAL SKILLS

Machine Learning: Deep Learning, Reinforcement Learning, Time-Series Classification, Predictive Modeling, Model Evaluation
Programming & Tools: Python, TensorFlow/Keras, scikit-learn, Pandas, NumPy, MATLAB, Java, C/C++/C#, VBA, Excel
Engineering Analytics: Data Preprocessing, Feature Engineering, Experiment Design, Data Visualization, Petrel, Petrosys

PROFESSIONAL EXPERIENCE

Petroleum Analyst — **Netherland Sewell & Associates, Inc. | Dallas, TX | Jul 2025-Present**

- Analyze engineering, geological, production, and economic datasets to evaluate oil and gas reserves and support data-driven technical decisions.
- Develop Python scripts and Excel models that streamline reserve-estimation, reporting, and forecasting workflows while improving accuracy and repeatability.
- Apply decline-curve analysis, volumetric calculations, and production forecasting to estimate hydrocarbon recovery and future performance.
- Translate complex analyses into technical reports and economic forecasts for multidisciplinary teams, clients, and regulatory stakeholders.
- Automated recurring data-management and reporting workflows using Python, Excel, and VBA, improving information accessibility, consistency, and repeatability across engineering operations.

IT Team Lead & Software Engineer — **Ozark Interests | Houston, TX | May 2022-Jul 2025**

- Designed and implemented a Python scheduling system that automated Excel-based assignment generation for more than 1,000 seasonal employees. Increased efficiency.
- Managed a company-wide technology inventory database and developed automation scripts to improve data quality, process efficiency, and issue tracking.
- Led technical support workflows and collaborated with operational teams to translate business requirements into reliable software and data solutions.

SELECTED MACHINE LEARNING PROJECTS

Electricity Theft Detection - 1D Convolutional Neural Network — **Jul 2026**

- Designed and tuned a 51,137-parameter 1D CNN to classify anomalous electricity-consumption behavior from 1,034-day SGCC smart-meter time series.
- Built a leakage-safe pipeline with training-only imputation, transformation, standardization, missing-value indicators, validation-based threshold selection, and sealed test evaluation.

Reinforcement Learning - MLB Batter Decision System — **Dec 2025**

- Built a MATLAB reinforcement-learning batter that learned at-bat decisions against MLB-level pitcher data, demonstrating sequential decision-making under uncertain conditions.

Robot Vision - Knight's Tour — **Apr 2025**

- Created a MATLAB-based 3D visualization system that computed and animated the Knight's Tour using trained movement models and image datasets.

Deep Neural Network - Pitch Velocity Prediction — **Nov 2024**

- Developed a Python neural network trained on thousands of pitcher statistics to predict pitch velocity across player profiles; performed data preparation, feature selection, training, and evaluation.

EDUCATION

M.S., Computer Engineering - Artificial Intelligence & Machine Learning — **Tarleton State University | 2024-2026 | GPA: 3.58**
B.S., Computer Science - Minor in Business — **Tarleton State University | 2020-2024 | GPA: 3.50**